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Human-Computer Interaction 1

2nd Semester SY 2025-2026

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Course and Year: BSCS 1

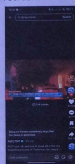
Date: 02/19/2026

Section: IDA6

SCORE

Activity: The Cognitive Audit

1. The Problem: A screenshot of the "bad" design with a brief description of the scenario.

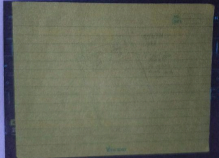




- TikTok, Instagram, and Facebook's "For You" Feed Algorithms & Content Preferences.
- A user attempts to "Reset" or "Fine-tune" their algorithm because the feed is "mixed up" with irrelevant or distressing content.
- The "Reset" function is buried under 4+ layers of nested menus (Settings > Content Preferences > Refresh Feed), creating a massive "Gulf of Execution".

2. The Cognitive Why: One sentence explaining the mental strain (e.g., "This creates high cognitive load because...").

- This design creates high cognitive/contextual overload by showcasing content in a chaotic order that stresses out the user.

3. The Fix: Your improved sketch or wireframe.



- This improved design sketch showcases an Algorithm Tuner dashboard, wherein you adjust the content preference settings to the right type of content that you wish to consume as to not overload you with different narratives all at the same time.

Activity: The Cognitive Audit
Goal: In pairs, identify a "Cognitive Design" failure in a common digital system, analyze why it fails the user's mind, and propose a standard solution.

Step-by-Step Instructions

- **Step 1: Form Your Design Pair:** Find a classmate. One of you will act as the "User" and the other as the "Observer/Designer."
- **Step 2: Choose a Target:** Pick a website or app you both use (e.g., your university portal, a government website, or a complex settings menu in an app).
- **Step 3: The "Think-Aloud" Test:**
 - The **User** attempts to complete a specific task (e.g., "Find the privacy settings and disable data sharing").
 - The **User** must "think aloud," describing what they are looking at and where they feel confused.
 - The **Observer** takes notes on where the "spotlight" of focus (Attention) is lost or where the "Gulf of Execution" (the gap between goal and action) appears.
- **Step 4: Analyze the Cognitive Impact:** Discuss which principle from Lesson 3.1 was violated:
 - **Memory:** Did the user have to recall a path rather than *recognizing* an icon?
 - **Perception:** Were grouped items unrelated (Gestalt failure)?
 - **Mental Model:** Did the system work differently than the user expected?
- **Step 5: Design the Solution:** Sketch a "Fixed Version" of that screen. Use **Visual Hierarchy** (size, color, placement) to make the task easier.

The Output (Submission)
 Submit a single one-page document (PDF or Image) containing:

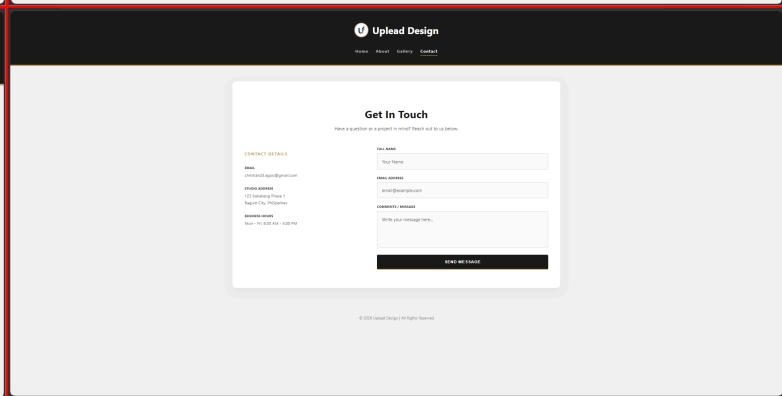
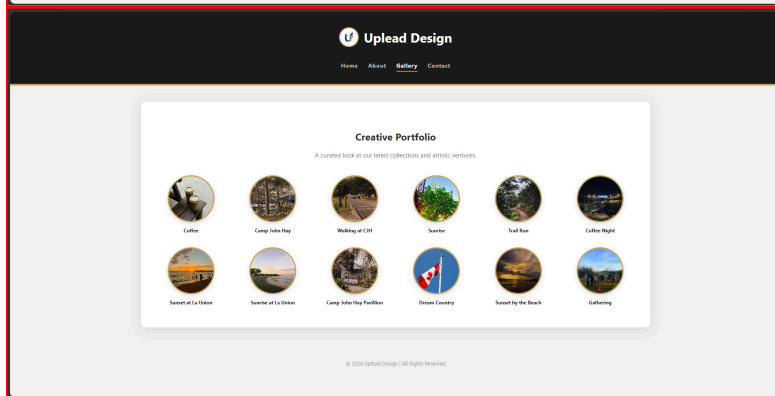
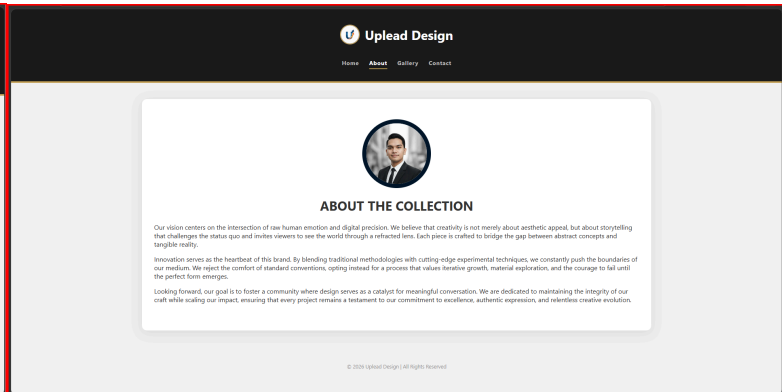
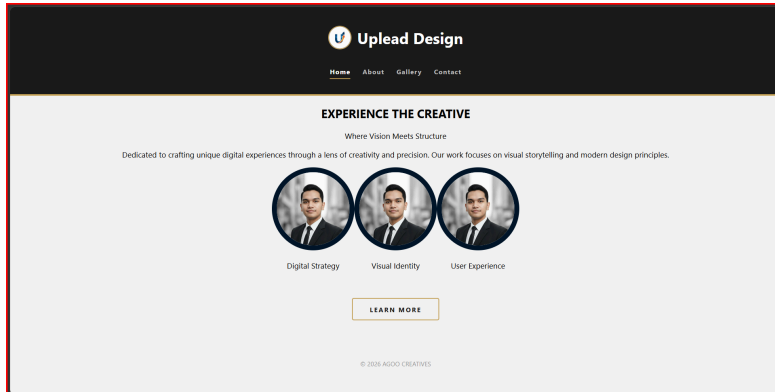
1. **The Problem:** A screenshot of the "bad" design with a brief description of the scenario.
2. **The Cognitive Why:** One sentence explaining the mental strain (e.g., "This creates high cognitive load because...").
3. **The Fix:** Your improved sketch or wireframe.
4. Be Ready to share it in the class.

Grading Rubric (Total: 20 Points)

Criteria	5 Points (Excellent)	3 Points (Developing)	1 Point (Needs Work)
Scenario Identification	Clearly identifies a specific task and a valid cognitive design flaw.	Identifies a flaw but the task is vague.	The chosen example isn't related to cognitive design.
Cognitive Analysis	Correctly uses terms like "Recall vs. Recognition," "Gulf of Execution," or "Mental Models."	Uses some HCI terms but applies them slightly incorrectly.	No mention of cognitive principles.
Proposed Solution	The fix clearly uses visual hierarchy or learnability principles to solve the problem.	The fix is better but doesn't address the specific cognitive issue.	The fix is just as confusing as the original.
Collaboration	Submission shows clear evidence of both student roles and "think-aloud" feedback.	Minimal evidence of interaction between partners.	Individual work with no peer feedback included.

[illegible]

MidTerms Lab Exam



MidTerms Lecture Exam

UNIVERSITY OF Baguio
SCHOOL OF INFORMATION TECHNOLOGY

HUMAN-COMPUTER INTERACTION/WEB DESIGN 1
2nd Semester SY 2022-2023
Midterm Examination

Name: JOANNA B. PASUAL Date: 06/01/23 Section: 007 Score: 35/40

GENERAL INSTRUCTIONS

1. Use blue or black permanent ink for answering. Entries containing erasures, corrections, or superimpositions will be considered invalid, and such answers will be marked as incorrect.
2. Mind your own test paper. Anyone caught cheating will automatically receive a score of zero (0) for the test and will be subject to disciplinary action in accordance with the provisions of the Student Handbook and other applicable university or school policies.
3. ALL gadgets must be set to silent mode and kept inside the bag at all times.
4. For this exam, you are the Lead UX Auditor for the mobile app "UniBook". For each of the four items, you must complete two parts:
 - The Audit (Analysis): Examine the "Problem UI". Identify the specific design failing using HCI terminology, and explain its impact on the user.
 - The Redesign (Application): Use the provided grid to sketch a visual solution that demonstrates an improvement in the specific HCI category being tested.
5. For items 1, 2, and 4, you can earn up to 10 points per item, with 5 points awarded for your Audit and 5 points for your Redesign sketch. Item 3 carries a higher weight of 20 points (split 10/10) because it demands a deeper mastery of Affective Computing and Persuasive Technology.

I. The Audit Scenario

You have been hired as a Lead UX Auditor for "UniBook", a mobile application designed to help university students book study rooms. However, user feedback indicates that the current interface is confusing, frustrating, and inefficient, leading to high abandonment rates. Your goal is to conduct a professional audit of the application's current design and propose a redesigned, user-centered interface.

Midterm Grading Rubric (UniBook HCI Exam)		
Mastery Level	Criteria for Item number 1, 2, and 4 (10 pts)	Criteria for Item number 3 (20 pts)
Mastery	10 pts: Complete audit with correct HCI terminology; sketch is functional and solves the problem.	20 pts: Superior use of Affective Computing and Persuasive Technology to resolve user frustration.
Developing	7 pts: Audit is mostly correct; sketch is functional but lacks professional strategic depth.	14 pts: Good attempt at empathy, missing key Persuasive Trigger elements.
Basic	5 pts: Vague audit; sketch is incomplete or fails to clearly address the identified issue.	10 pts: Friendly error message provided but lacks a strategic UI solution/success path.
Unsatisfactory	0 pts: No audit or sketch provided.	0 pts: No audit or sketch provided.

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1. Cognitive Load (Hick's Law): The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making.

Task 1: The Audit (5 pts): Explain how this interface violates Hick's Law. Why does a high number of choices increase cognitive load and decision time?

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

Task 2: The Redesign (5 pts): Use the grid to sketch a "Chunked" menu design that reduces visual clutter.

Problem UI: UniBook

Solution Sketching Area

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

2. Social Presence (Co-presence): Users feel isolated when booking rooms, as the app provides no visibility into whether friends or study groups are already using nearby facilities.

Task 1: The Audit (5 pts): Analyze the lack of Co-presence in the current UI. How does this lack of social awareness impact the user's motivation to use the app?

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

Task 2: The Redesign (5 pts): Use the grid to sketch a "Chunked" menu design that reduces visual clutter.

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Task 2: The Redesign (5 pts): Use the grid to sketch a "Presence/Sync" feature that allows students to see where their peers are studying.

Problem UI: UniBook

Solution Sketching Area

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

3. Affective Computing (Emotional Design): When a student attempts to book an unavailable room, the app displays a cold "Error 404: Invalid Selection." Students report feeling frustrated and tend to quit.

Task 1: The Audit (10 pts): Audit this message using Affective Computing principles. Why is this error message a failure in emotional design?

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

Task 2: The Redesign (5 pts): Use the grid to sketch a "Presence/Sync" feature that allows students to see where their peers are studying.

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Task 2: The Redesign (10 pts): Rewrite the message as a Persuasive Trigger.

Problem UI: UniBook

Solution Sketching Area

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

4. Information Architecture (IA Fatigue): The UniBook app requires users to navigate 7 separate screens to finalize a single room booking. This high number of interactions causes "IA fatigue".

Task 1: The Audit (5 pts): Identify the flaw in the current Information Architecture (IA). Why is a 7-step process detrimental to user conversion?

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

Task 2: The Redesign (5 pts): Use the grid to sketch a "Presence/Sync" feature that allows students to see where their peers are studying.

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Task 2: The Redesign (5 pts): Tooling: List 2 industry-standard prototyping tools (e.g., Figma, Adobe XD) that professionals use to build and test efficient workflows. Use the grid to sketch a "One-Click" or 3-step booking flow that reduces the friction of the original process.

Problem UI: UniBook

Solution Sketching Area

Handwritten notes: The dashboard displays 15 unorganized navigation icons, leading to student confusion and slow decision-making. Hick's Law states that the time it takes to make a decision increases with the number of choices. A high number of choices increases cognitive load and decision time.

Prepared by: Joana Rosa B. Pasual, Instructor

Reviewed by: Divine L. Aguilar-Agudong, Program Chair, IT; Cherrin L. Amazan, Program Chair, CS/ACT

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